$$\partial_t U + \partial_x F(U) = 0$$

$$U = (\rho, m, E)^{\mathsf{T}}, \quad F(U) = \begin{pmatrix} m \\ m^2/\rho + p \\ m(E+p)/\rho \end{pmatrix}$$

$$\bar{U}_i^{n+1} = \bar{U}_i^n - \frac{\Delta t}{\Delta x}\left(\widehat{F}_{i+1/2} - \widehat{F}_{i-1/2}\right)$$

$$\mathcal{R}_{V2} = \mathcal{R}_T + G\lambda\left(\mathcal{R}_{NN} - \mathcal{R}_T\right)$$

$$p = (\gamma - 1)\left(E - \frac{m^2}{2\rho}\right)$$

$$\rho > 0, \qquad p > 0$$

$$F^{\pm}(U) = \frac{1}{2}\left(F(U) \pm \alpha U\right)$$

$$\mathrm{CFL} \sim \frac{\alpha \triangle t}{\triangle x}$$

$$\bar{U}_i^{n+1} = \bar{U}_i^n - \frac{\Delta t}{\Delta x}\left(\widehat{F}_{i+1/2} - \widehat{F}_{i-1/2}\right)$$

$$\Delta M_{\text{repair}} = \Delta x \sum_i (\rho_i^{after} - \rho_i^{before})$$

$$\overline{F(U)} \neq F(\bar{U})$$

$$\widehat{F}^{limited} = \widehat{F}^{LO} + \theta\left(\widehat{F}^{NN} - \widehat{F}^{LO}\right)$$

$$\mathscr{R} = \partial_t \hat{U} + \partial_x F(\hat{U})$$

$$\lambda_j = v - c,\ v,\ v + c, \qquad c = \sqrt{\gamma p/\rho}$$

$$g^{\pm}_{i+1/2,k} = \frac{1}{2} L_{i+1/2} \left(F(U_k) \pm a_{i+1/2} U_k \right)$$

$$\widehat{F}_{i+1/2} = R_{i+1/2}\left[\mathcal{R}(g^{+}) + \mathcal{R}(g^{-})\right]$$

$$U^{(1)} = U^n - \Delta t D(U^n)$$

$$U^{(2)} = \frac{3}{4}U^n + \frac{1}{4}\left[U^{(1)} - \Delta t D(U^{(1)})\right]$$

$$U^{n+1} = \frac{1}{3}U^n + \frac{2}{3}\left[U^{(2)} - \Delta t D(U^{(2)})\right]$$

$$\rho > 0, \qquad E - \frac{m^2}{2\rho} > 0$$

$$\alpha_k = \frac{d_k}{(\epsilon + \beta_k)^2}, \qquad \omega_k = \frac{\alpha_k}{\Sigma_j \, \alpha_j}$$

$$\left(10^{-6}\right)^2 = 10^{-12} \gg 10^{-38}$$

$$U_h|_{I_i} = \sum_{\ell=0}^{q} U_{i,\ell}\varphi_\ell(x)$$

$$\widehat{U} = K_h * U_h, \qquad \int K = 1, \quad \int x^r K(x)\,dx = 0$$

$$\mathscr{R} = \partial_t \hat{U} + \partial_x F(\hat{U})$$